

POP

Full Stack Builder Assignment

Assignment Overview

You are tasked with building a financial insights product for users based on their UPI transaction data. This assignment evaluates your full stack capabilities, product sense, design skills, and native development expertise.

Context

Assume you are a full stack builder at POP. Your objective is to build a product that provides meaningful financial insights to users on their UPI transactions, enabling them to understand their spending patterns and financial behavior better.

You have been provided with [synthetic transaction data](#) for 150 different users (~20K transactions total) as a reference dataset. This data should inform your design insights and help you validate your implementation.

Requirements

Product & User Understanding

- Understand the users: Analyze the provided transaction data to identify user personas, spending behaviors, and key financial patterns.
- Design insights: Determine what financial insights would be most valuable to these users. Examples include spending by category, monthly trends, recurring expenses, highest spenders, savings potential, etc.
- POP-specific angle: Frame these insights within POP's value proposition and positioning as a UPI-based payments app.

Design (UI/UX)

- Create design files with a clean, intuitive UI for displaying financial insights in line with POP's design. Reference design file is shared [here](#).
- Design screens for key user journeys
- Ensure visual hierarchy, readability, and usability optimized for mobile screens.
- Use Figma (or equivalent) to create high-fidelity mockups and design specs.

Development

Primary: Mobile-First Web App

- Build a responsive, mobile-first web application using any modern framework (React, Vue, Svelte, Next.js, etc.).
- Implement backend APIs for transaction data, insights computation, and user authentication.
- Set up a database (relational or NoSQL) to store transaction data and user information.

- Deploy the web app to a live hosting service (Vercel, Railway, Netlify, AWS, etc.) so it can be immediately accessed and tested on a phone browser.
- Ensure polished performance and usability on mobile browsers (touch interactions, responsive design, etc.).

Secondary: Native Implementation

- **Choose one platform:** Build one screen from your web app as a native Android (Kotlin + Jetpack Compose) or iOS (SwiftUI) implementation.
- Mock data is acceptable—the goal is to demonstrate your native development skills and code quality.
- This showcases your ability to implement designs across platforms and highlights your native coding proficiency.

Deliverables

Your submission must include the following:

Deliverable	Description
Live Web App	A deployed, fully functional mobile-first web app accessible via a public URL. Include a Usage Guide (in the README or within the app) documenting features, how to use it, and any backend setup instructions. The app should be immediately testable on a phone browser—no build process required.
Native Screen	A native Android (Kotlin + Jetpack Compose) or iOS (SwiftUI) implementation of one screen from your web app. Use mock data if needed. Include build instructions (APK for Android, Xcode for iOS) in your repository.
Design Files	Figma (or equivalent) design file(s) containing UI designs for all key screens. Include annotations explaining design decisions, typography, color schemes, and component specifications.
Codebase	Complete, well-documented source code for the entire project (web frontend + backend + native app). Push to a public GitHub repository with clear README instructions for running and testing locally, deploying the web app, and building the native app.

Evaluation Criteria

Product Sense

- Understanding of user needs and financial behavior
- Relevance and value of chosen insights
- Alignment with POP's positioning and user base
- Documentation of product thinking and decision-making

Design Skills

- Visual hierarchy and information architecture
- Mobile-first design approach and responsiveness
- Consistency in design system and component usage
- Quality of visual assets and design documentation

Development Skills

Web Development

- Code quality, architecture, and design patterns
- Backend API design and data processing logic
- Database design and schema
- Frontend performance, responsiveness, and user experience

Native Development

- Native code quality and proper use of platform APIs
- Understanding of Jetpack Compose/SwiftUI patterns and best practices
- Ability to implement design fidelity in native code

Timeline & Scope

Expected completion: 8–10 hours of focused work. This is a contained sprint to evaluate your ability to ship a complete product in a short timeframe. Scope accordingly—prioritize a polished, end-to-end experience over feature breadth.

If you need more time, please communicate early with your recruiting contact.

Submission Instructions

1. **GitHub Repository:** Push your complete codebase (web + backend + native) to a public GitHub repository with clear README files for each component. Include build/deployment instructions.
2. **Live Web App:** Share the public URL of your deployed web app in the main README. It should be immediately accessible and testable on a phone browser.
3. **Native App:** Include source code and build instructions (Android: APK file; iOS: Xcode project). Screenshots or a demo video are optional but appreciated.
4. **Design Files:** Share a link to your Figma (or equivalent) design file. Ensure the file is publicly accessible for review.
5. **Documentation:** Include a brief write-up in the README outlining your product thinking, user insights, design rationale, tech stack choices, and any technical decisions.

Additional Notes

- **Tech Stack:** Use any modern web framework, backend (Node.js, Python, Go, Rust, etc.), and database (PostgreSQL, MongoDB, Firebase, etc.). For native: Kotlin + Jetpack Compose (Android) or SwiftUI (iOS). Choose based on your expertise.
- **Deployment:** Use any hosting provider (Vercel, Netlify, Railway, AWS, Fly.io, etc.). Free tiers are fine. The key is that the app must be live and testable immediately.
- **Data Privacy:** Treat user financial data with care.

- **Testing Data:** The provided synthetic dataset is for reference. You may use it as seed data for your database or generate additional realistic test data.
- **Code Organization:** Ensure your code is clean, well-structured, and easily reviewed. Use proper commit messages and document key architecture decisions.
- **Scope & Completeness:** With 8–10 hours, ship something complete and polished rather than many half-baked features. A tight, well-executed MVP is better than feature bloat.

References:

- [Dataset](#)
- [Design File](#)

Questions?

If you have any questions about the assignment, please reach out to your recruiting contact. We're excited to see what you build!